

SCT_DA_3

HR Employee Attrition Dashboard Final Report

SkillCraft Technology Data Analytics Internship – Task 03

Executive Summary

This project focuses on developing an **interactive HR Employee Attrition Dashboard** using **Power BI** to analyze employee turnover patterns and uncover the major factors contributing to workforce attrition.

Employee retention is a critical business challenge, and understanding why employees leave can help organizations improve productivity, reduce hiring costs, and strengthen workforce stability.

Using the IBM HR Employee Attrition dataset, this dashboard provides data-driven insights into employee departures by analyzing demographics, job roles, compensation, overtime, and satisfaction metrics.

Core Business Question:

Why are employees leaving?

Project Objectives

The primary objectives of this project were:

- Build a professional interactive dashboard using Power BI
 - Analyze employee attrition across multiple dimensions
 - Enable dynamic filtering for deeper exploration
 - Identify major drivers of employee turnover
 - Provide actionable HR recommendations
 - Improve strategic decision-making through data visualization
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Dataset Information

Dataset Used:

IBM HR Analytics Employee Attrition Dataset

Dataset Features:

- Total Employees: 1,470
- Employee Age
- Department
- Gender
- Job Role
- Monthly Income
- Attrition Status
- Overtime
- Job Satisfaction
- Education Field
- Marital Status
- Years at Company
- Business Travel

Tools & Technologies Used

Tool	Purpose
Power BI Desktop	Dashboard creation
Power Query	Data cleaning and transformation
DAX	KPI calculations and measures
Microsoft Excel	Dataset preparation
GitHub	Project repository and portfolio showcase

Data Cleaning & Preparation

Several preprocessing steps were performed to ensure data accuracy and analytical quality:

Data Cleaning Steps:

- Removed redundant columns:
 - EmployeeCount
 - Over18
 - StandardHours
- Verified and corrected data types

- Checked for null values and inconsistencies
- Created custom columns:
 - Age Group
 - Income Group
- Structured data model for efficient reporting

Custom Segments Created:

Age Groups:

- 18–25
- 26–35
- 36–45
- 46+

Income Groups:

- Low Income
 - Medium Income
 - High Income
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KPI Metrics Developed

The dashboard includes the following core metrics:

- **Total Employees**
- **Employees Left**
- **Attrition Rate**
- **Average Monthly Income**
- **Average Job Satisfaction**
- **Average Years at Company**

These KPIs provide an executive-level overview of organizational workforce health.

Dashboard Features

Interactive Filters:

Users can filter dashboard insights by:

- Department
- Age Group

- Gender
 - Job Role
 - Marital Status
 - Education Field
 - Business Travel
 - Income Group
 - Overtime Status
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Key Visualizations:

1. Attrition by Department

Identifies departments with the highest turnover.

2. Attrition by Age Group

Highlights vulnerable employee demographics.

3. Attrition by Income Group

Examines salary-based retention challenges.

4. Overtime Impact Analysis

Measures overtime's role in employee exits.

5. Job Satisfaction Analysis

Shows correlation between satisfaction and attrition.

6. Gender-wise Attrition

Analyzes workforce turnover by gender.

7. Years at Company Trend

Tracks attrition relative to employee tenure.

Key Findings

Based on dashboard analysis, several critical insights were identified:

Major Attrition Drivers:

1. Overtime

Employees working overtime show significantly higher attrition rates.

2. Age Group 26–35

This demographic experiences the highest turnover.

3. Low Monthly Income

Employees in lower salary brackets are more likely to resign.

4. Job Dissatisfaction

Low job satisfaction strongly correlates with employee exits.

5. Early Career Employees

Employees with fewer years at the company are more vulnerable to leaving.

Business Recommendations

To improve employee retention, the following strategies are recommended:

Strategic Actions:

Reduce Overtime Burden

- Improve work-life balance
- Optimize staffing levels
- Reduce burnout

Improve Compensation Structures

- Review salary competitiveness
- Focus on low-income roles
- Introduce incentive programs

Strengthen Employee Engagement

- Increase job satisfaction initiatives
- Career development opportunities
- Recognition programs

Focus on Younger Workforce

- Develop retention strategies for 26–35 age group
- Enhance onboarding and mentoring

Department-Specific HR Planning

- Prioritize high-attrition departments
- Build targeted interventions

Dashboard Design & UI Strategy

A custom professional dark-themed dashboard interface was created to:

- Improve readability
- Enhance executive presentation
- Support recruiter portfolio value
- Deliver modern business intelligence aesthetics

Design Features:

- Dark navy background
 - Orange accent highlights
 - Rounded KPI cards
 - Structured chart panels
 - Insights & Recommendations sections
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Challenges Faced

During project development, several challenges were addressed:

- Data cleaning and structuring
- Designing effective KPI calculations
- Creating professional UI layout
- Ensuring dashboard interactivity
- Optimizing visual storytelling

These challenges strengthened practical Power BI and analytical problem-solving skills.

Skills Demonstrated

This project showcases proficiency in:

- Data Cleaning
- Data Transformation
- DAX Calculations
- Power BI Dashboard Development
- HR Analytics
- Interactive Reporting
- Business Intelligence
- Strategic Analysis
- Visual Storytelling

- GitHub Portfolio Development
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Project Outcome

The final dashboard successfully delivers:

- ✓ **Clear attrition trend analysis**
 - ✓ **Executive-level KPI overview**
 - ✓ **Interactive business intelligence tools**
 - ✓ **Strategic recommendations**
 - ✓ **Professional portfolio-quality presentation**
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Conclusion

The **HR Employee Attrition Dashboard** provides a powerful analytical solution for understanding workforce turnover and improving HR strategy.

By transforming raw employee data into actionable insights, this project demonstrates how business intelligence tools can support better organizational decision-making.

This project not only fulfills the internship task requirements but also serves as a strong portfolio project showcasing advanced capabilities in:

- Power BI
 - Data Analytics
 - HR Business Intelligence
 - Dashboard Design
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Future Enhancements

Potential future improvements include:

- Predictive attrition modeling
 - Machine learning integration
 - Real-time HR monitoring
 - Advanced decomposition trees
 - AI-based workforce forecasting
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Final Note

This project demonstrates the practical application of data analytics, visualization, and business intelligence to solve real-world HR challenges through professional dashboard development.